

The Fermionic Matter Sub-Programme

Presentation Note 6

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Abstract

The fermionic matter sub-programme of the Cosmochrony corpus addresses a single central question: where do fermions, chirality, hypercharge, and three generations come from? The answer is that they are not postulated as external fields: they arise as the spinorial face of the admissible Weil module V_ρ , once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric established in the geometric branch. The structural chain is:

$$\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2, \mathbb{R}) \Rightarrow \mathrm{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3, 1) \simeq \mathrm{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi.$$

Three modular structural results are established in Q14: (A) the admissible spinor bundle $\mathcal{S}_\Pi$ reproduces the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ electroweak bundle structure from tensor functors of $\mathcal{S}_\Pi$; (B) the projected Dirac operator $D_{\Pi, g, A}$ contains a canonical projective endomorphism E_Π that enforces the $V - A$ chiral structure (established via the spinorial BI lift theorem of Q14) and whose γ_5 -weighted a_4 coefficient imposes anomaly-cancellation constraints making hypercharge a spectral datum; (C) the saturation invariant $\sigma_c(n_3) = 3$ has a spinorial multiplicity reading yielding a gauge-singlet three-generation factor C_{gen}^3 . The colour-coupled quark sector is unconditional at the pointwise level ($[\mathrm{H}\text{-color}]_{\mathrm{pointwise}}$ established in O31). Open deliverables are the explicit form of E_Π and the Yukawa sector.

Keywords. fermionic matter; admissible spinor bundle; metaplectic lift; chirality; V-A structure; hypercharge rigidity; anomaly cancellation; three generations; projected Dirac operator; Weil representation; non-injective projection; presentation note.

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1 Motivation and Central Question

The fermionic matter sub-programme answers the following question.

Central question. The bosonic spectral stratification — gravity at a_2 , Yang–Mills at a_4 — derives the bosonic sector of the Standard Model from the same admissible spectral functional. Fermions, chirality, hypercharge assignments, and the three-generation structure are not covered by the bosonic synthesis. *Do these fermionic structures also arise as forced consequences of the admissible Weil fibre, or must they be introduced as additional postulates?*

The answer established in Q14 [2] is: they are forced. Fermions arise as the spinorial face of the Weil module V_ρ after Lorentzian complexification of its metaplectic lift. Chirality and the $V - A$ structure arise from the projective endomorphism E_Π of the projected Dirac operator, via the spinorial lift of BI parity established in Q14. Hypercharge is not a free parameter: it is constrained by spectral anomaly cancellation. Three generations arise from the spinorial multiplicity reading of the same rank-three invariant $\sigma_c(n_3) = 3$ that yields the three spatial directions in the geometric branch.

This sub-programme is the first paper of the Q-series that depends simultaneously on all four layers of the programme: the admissible Weil fibre structure (Notes 1 and 5), the closed Lorentzian geometry (Note 2), the gauge bundle (Note 3), and the gauge–gravity synthesis (Q13 [3], to be summarised in Presentation Note 8).

Remark 1 (Scope: identification only, not mass spectrum). This sub-programme closes the *identification* of the fermionic sector: spinor bundle, chirality, hypercharge quantum numbers, and generation factor. It does not derive the mass spectrum, the Yukawa couplings, or the CKM/PMNS mixing matrices. Those require the explicit construction of E_Π as an operator on the spectral space, and the integration of the projected Yukawa Lagrangian $\mathcal{L}_Y$, both of which are listed as open deliverables.

Remark 2 (Vocabulary note). Throughout this note, *admissibility constraint* replaces the earlier “relaxation” vocabulary. The substrate χ is static; fermions are not excitations of χ but the spinorial interpretation of the admissible fibre after geometric complexification.

2 Position in the Cosmochrony Programme

The fermionic matter sub-programme sits at the apex of Branch III. It depends on all five preceding sub-programmes and provides the fermionic matter content consumed by any downstream phenomenological application.

Inputs from upstream:

- $F_n \simeq V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$, canonical symplectic structure of V_ρ , and the parity involution $c \leftrightarrow q - c$ (Notes 1 and 5).
- Effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ and effective base manifold $M_{\text{eff}} = \mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ (Note 2, Q5b–Q11).
- Admissible principal bundle $P_{G_\Pi}(M_{\text{eff}}, G_\Pi)$ and gauge connection A (Note 3, Q6a).
- $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ gauge group (Note 3).
- Spectral entropy functional $S_\Pi[g, A]$ and the a_4 variation (Note 4/Q12, Q13 [3]).
- $H_{\text{eff}} = \text{Sym}^2(V_\rho)$, three spatial directions from $\text{Im } \mathbb{H}$, $\sigma_c(n_3) = 3$ (Note 2, Q7 [4]).

3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following structural chain.

$$\begin{aligned} \Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \text{Mp}(2, \mathbb{R}) \Rightarrow \text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi \quad (1) \\ \Rightarrow \underbrace{\text{Sym}^2(\mathcal{S}_\Pi) \Rightarrow \text{ad}_{\mathbb{C}}(P_\Pi)}_{\text{SU}(2)_L}, \quad \underbrace{\wedge^2(\mathcal{S}_\Pi) = L_Y}_{\text{U}(1)_Y}, \quad \underbrace{E_\Pi \text{ left-admissible}}_{V-A}, \quad \underbrace{\sigma_c(n_3) = 3 \rightarrow C_{\text{gen}}^3}_{3 \text{ generations}}. \end{aligned}$$

Three distinct mechanisms contribute.

1. *Lorentzian complexification forces the spin group.* The Weil module V_ρ carries a canonical symplectic structure; its automorphisms lift to $\text{Mp}(2, \mathbb{R})$. The effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Lorentzian) forces the Lie algebra complexification $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, whose Lorentzian integration is $\text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C})$. The spinor bundle $\mathcal{S}_\Pi$ is the associated fundamental bundle.
2. *Tensor functors of $\mathcal{S}_\Pi$ reproduce the electroweak bundle.* $\text{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C})$: the symmetric sector gives the complex adjoint, from which the compact real form selects the $\text{SU}(2)_L$ gauge sector. $\wedge^2(\mathbb{C}^2) \simeq \det(\mathbb{C}^2)$: the determinant line L_Y supports the $\text{U}(1)_Y$ factor. No independent electroweak matter bundle is introduced.
3. *The projective endomorphism E_Π enforces chirality and hypercharge.* The projected Dirac operator satisfies the Lichnerowicz-type identity

$$D_{\Pi, g, A}^2 = -(\nabla^{S, A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu \gamma^\nu F_{\mu\nu}^a T_R^a + E_\Pi.$$

By the spinorial lift of BI parity, established in Q14 [2], E_Π is left-admissible ($P_R E_\Pi P_R = 0$), giving the $V - A$ structure. $\text{U}(1)_Y$ invariance of $S_\Pi[g, A, D_{\Pi, g, A}]$ imposes the anomaly-cancellation trace constraint $\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$, fixing hypercharge as a spectral coherence datum.

4 Constituent Papers

The sub-programme has a single primary paper.

4.1 Q14: Fermionic matter and chirality from projective Dirac admissibility

Q14 [2] is the sole constituent paper. It is organised around three independent but mutually compatible structural results.

Theorem A: Spinorial electroweak bundle (structural; unconditional). The admissible Weil module V_ρ induces, via the metaplectic lift and the Lorentzian complexification forced by $g^{\mu\nu} = 2\eta^{\mu\nu}$, an admissible spinor bundle $\mathcal{S}_\Pi$. Its tensor sectors satisfy:

$$\text{Sym}^2(\mathcal{S}_\Pi) \simeq \text{ad}_{\mathbb{C}}(P_\Pi), \quad \wedge^2(\mathcal{S}_\Pi) = L_Y.$$

After selection of the compact real form, these define the $\text{SU}(2)_L \times \text{U}(1)_Y$ electroweak bundle data without any additional input.

Theorem B: Chiral selection and hypercharge rigidity (structural; unconditional). The projected Dirac operator $D_{\Pi, g, A}$ has the Lichnerowicz-type square above. By the spinorial lift of BI parity, established in Q14 (the antilinear $J_\Pi = HK$ maps right-chiral modes to non-admissible

directions), E_Π is left-admissible: $P_R E_\Pi P_R = 0$. This is the $V - A$ structure. $U(1)_Y$ invariance of the projected spectral functional then imposes, at the a_4 level:

$$\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0 \quad \forall x \in M_{\text{eff}},$$

where A_Π is the γ_5 -weighted a_4 Seeley–DeWitt density of $D_{\Pi,g,A}^2$. This constraint is not postulated: it is the chiral consistency condition of the same a_4 level that yields Yang–Mills dynamics in the bosonic variation. The hypercharge weights Y_R are thereby not free parameters but spectral coherence data.

Theorem C: Generation multiplicity (structural; unconditional). The saturation invariant $\sigma_c(n_3) = 3$ has two functorially distinct readings of the same rank-three admissible invariant. In the geometric branch (Note 2), it yields three effective spatial directions. In the spinorial branch, the spinorial multiplicity functor maps it to a gauge-singlet generation space:

$$C_{\text{gen}}^3 \subset \ker(\text{ad}_{\text{SU}(2)} \oplus Y).$$

The identity $\mathbb{R}_{\text{space}}^3 \neq C_{\text{gen}}^3$ is established explicitly (Q14 §5): the spatial and generation spaces are distinct functorial images of the same admissible invariant. Three generations are forced; their physical distinction requires the explicit construction of E_Π (open).

Colour sector (unconditional at the pointwise level). The colour-coupled quark sector $\mathcal{S}_\Pi \otimes V_{\text{color}}$ is obtained by tensoring the admissible spinor bundle with the colour module V_{color} . The status of this sector inherits the status of [H-color] from Notes 1 and 3: established unconditionally on the colour-adapted graph, and on the standard graph at the rank, averaged, and effective-exponent levels, with [H-color]_{pointwise} proved analytically (O31, exact equality of BFS capacity profiles at finite q). The only residual component is the extension beyond the standard Cayley graph to general generating sets, which is non-blocking for the quark bundle.

Status of the spinorial BI lift. The spinorial lift of BI parity to the spinor bundle $\mathcal{S}_\Pi$ is established as a theorem in Q14: the BI parity involution $c \leftrightarrow q - c$, proved at the scalar Weil level in O18 and BornInfeld, lifts to the antilinear endomorphism $J_\Pi = HK$ satisfying $J_\Pi^2 = -1$ and $J_\Pi \gamma_5 = -\gamma_5 J_\Pi$, dressing the parity by the weight grading $H = 2J_3$. Chirality and $V - A$ in Q14 are therefore unconditional, their only residual dependence being the axiomatic chain A1–A4 [1] and the upstream BI parity datum (O30, U1).

Table 1: Summary of Q14 results by theorem. Status: P = proved; S = structural.

Result	Central output	Status
Theorem A	Spinor bundle $\mathcal{S}_\Pi$; EW bundle from tensor functors	S (uncond.)
Theorem B ($V - A$)	$P_R E_\Pi P_R = 0$; $V - A$ structure	S (uncond.)
Theorem B (hypercharge)	Y_R from anomaly-cancellation trace; not free	S (uncond.)
Theorem C	C_{gen}^3 ; three generations as gauge-singlet factor	S (uncond.)
Colour sector	$\mathcal{S}_\Pi \otimes V_{\text{color}}$; quark bundle	S (uncond., pointwise)

5 Inputs from Upstream Results

1. $F_n \simeq V_\rho$ **and symplectic structure** (Notes 1, 5): the identification of the admissible fibre as the Weil representation is the starting point of the metaplectic lift. Without this, the spinor bundle $\mathcal{S}_\Pi$ has no canonical definition.
2. **Lorentzian metric** $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Note 2, Q5b–Q11): the Lorentzian signature forces the complexification $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$. Without the Lorentzian metric, the spin group would remain $\text{Sp}(2, \mathbb{R})$ and no Dirac spinors would arise.

3. **Gauge bundle and** $SU(2)_L \times U(1)_Y \times SU(3)$ (Note 3, Q6a): the admissible principal bundle and gauge connection A enter the spin–gauge connection $\nabla^{S,A}$ of the projected Dirac operator.
4. **a_4 Yang–Mills variation** (Q12 [5], Q13 [3]): the spectral anomaly cancellation of Theorem B uses the same a_4 -level structure that yields Yang–Mills dynamics in the bosonic sector. The fermionic a_4 coefficient is a structural extension of the bosonic one. The gauge–gravity synthesis of Q13 is the scientific dependency here; its presentation-level treatment will appear in Presentation Note 8.
5. $\sigma_c(n_3) = 3$ (Note 1, O23): the rank-three saturation invariant provides the spinorial multiplicity input for Theorem C.
6. **BI parity involution** (BornInfeld, O18): the scalar BI parity $c \leftrightarrow q - c$ is the input from which the spinorial BI lift is derived; its derivation at the scalar level is complete.

6 Outputs to Downstream Branches

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
Admissible spinor bundle $\mathcal{S}_\Pi$	Phenomenology, Q-series	S
$SU(2)_L \times U(1)_Y$ from tensor functors	SM phenomenology	S
$V - A$ chiral structure	Weak interaction pheno	S (uncond.)
Hypercharge as spectral datum	SM charge assignment	S (uncond.)
Three-generation factor C_{gen}^3	Mass/mixing notes	S
Quark bundle $\mathcal{S}_\Pi \otimes V_{\text{color}}$	QCD sector	S (uncond., pointwise)
E_Π as projective endomorphism	Yukawa sector (open)	S (open)

7 Status of Results

7.1 Structural results (unconditional)

1. *Admissible spinor bundle* (Theorem A): $\mathcal{S}_\Pi$ is induced from V_ρ via the metaplectic lift and Lorentzian complexification. The construction is algebraic and functorial; it depends on the Lorentzian closure of Q5b–Q11 but not on the spinorial BI lift.
2. $SU(2)_L$ from $\text{Sym}^2(\mathcal{S}_\Pi)$ (Theorem A): the symmetric tensor sector gives the complex adjoint $\text{ad}_{\mathbb{C}}(P_\Pi) \simeq \mathfrak{sl}_2(\mathbb{C})$; after compact real-form selection, the $SU(2)_L$ gauge sector is identified.
3. $U(1)_Y$ from $\wedge^2(\mathcal{S}_\Pi)$ (Theorem A): the determinant line $L_Y = \wedge^2(\mathcal{S}_\Pi)$ is the geometric support of the hypercharge factor.
4. *Generation multiplicity* (Theorem C): $C_{\text{gen}}^3 \subset \ker(\text{ad}_{SU(2)} \oplus Y)$ is a gauge-singlet three-generation space arising from the spinorial multiplicity reading of $\sigma_c(n_3) = 3$. The spatial and generation readings of the rank-three invariant are distinct ($\mathbb{R}_{\text{space}}^3 \neq C_{\text{gen}}^3$; Q14 §5).

7.2 Further structural results (unconditional)

1. *Left-admissibility of E_Π* (Theorem B): $P_R E_\Pi P_R = 0$ follows from the spinorial BI lift (Q14), now established.
2. *$V - A$ chiral structure* (Theorem B): the condition $P_R E_\Pi P_R = 0$ selects the left-chiral admissible projection $P_L \mathcal{S}_\Pi$, giving the $V - A$ coupling of weak interactions.
3. *Hypercharge rigidity* (Theorem B): the anomaly-cancellation trace condition $\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$ constrains Y_R to spectral coherence data; hypercharge is not a free parameter of the framework.
4. *Quark bundle* (§6): $\mathcal{S}_\Pi \otimes V_{\text{color}}$ is structural and unconditional at the pointwise level: the spinorial BI lift is established (Q14) and $[\text{H-color}]_{\text{pointwise}}$ is proved analytically (O31).

7.3 Open problems

1. *Explicit form of E_Π* . E_Π is identified as the zero-order endomorphism of the projected Dirac square, the “internal spectral residue of non-injective projection.” Its explicit form as an operator on the spectral space — analogous to the finite internal Dirac operator of Connes–Chamseddine spectral triples — is not yet constructed. Without the explicit E_Π , the three-generation distinction and the Yukawa sector cannot be derived.
2. *Normalisation of $\mathcal{L}_Y$* . The Yukawa Lagrangian $\mathcal{L}_Y$ is identified as arising from the trace $\text{Tr}_{\mathcal{S}_\Pi}[E_\Pi^2]$ variation. Its integral normalisation and the derivation of the mass matrix from the spectral data of E_Π are open.
3. $[\text{H-color}]_{\text{pointwise}}$ *beyond the standard graph*. Inherited from Notes 1 and 3. $[\text{H-color}]_{\text{pointwise}}$ is proved on the standard Cayley graph (O31); its extension to general generating sets is the only residual, and is non-blocking for the quark bundle, which is already established at the pointwise level.

8 Open Deliverables

The derivation of the spinorial BI lift, formerly the first open deliverable, is now established as a theorem in Q14 [2]: the antilinear lift $J_\Pi = HK$ dresses the BI parity by the weight grading, making Theorems B and C unconditional. Two open deliverables remain.

Open deliverable 1: Explicit E_Π and the Yukawa sector. The mass spectrum of charged leptons and quarks, and the CKM/PMNS mixing matrices, require the explicit spectral construction of E_Π as an operator on the admissible Hilbert space. This requires integrating the projected Yukawa Lagrangian $\mathcal{L}_Y$ and extracting the mass matrix from the eigenvalues of E_Π^2 . This is the primary open problem of the fermionic sector.

Open deliverable 2: Full matter content in Lorentzian signature. The complete fermionic content $S_\Pi^{\text{matter}} = (\mathcal{S}_\Pi \oplus (\mathcal{S}_\Pi \otimes V_{\text{color}})) \otimes C_{\text{gen}}^3$ in full Lorentzian signature, with explicit gauge couplings and three-generation Yukawa structure, is the downstream target. It requires the resolution of Deliverable 1 above, the colour sector being already established at the pointwise level (O31).

9 Conclusion

The fermionic matter sub-programme establishes that fermions, chirality, hypercharge, and three generations are not external additions to the Cosmochrony framework. They are the spinorial face of the admissible Weil module V_ρ , revealed through the Lorentzian complexification of its metaplectic structure.

The electroweak bundle $SU(2)_L \times U(1)_Y$ arises from the tensor functors of the admissible spinor bundle $\mathcal{S}_\Pi$ (Theorem A, unconditional). The $V - A$ chiral structure and hypercharge rigidity arise from the projective endomorphism E_Π (Theorem B, unconditional). Three generations arise from the spinorial multiplicity reading of $\sigma_c(n_3) = 3$ (Theorem C, unconditional). The colour-coupled quark sector follows by tensoring with V_{color} , unconditional at the pointwise level (O31).

The remaining open components are the explicit construction of E_Π and the derivation of the Yukawa sector and mass spectrum.

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